

Problem Definition

The objective of this project is to build a predictive machine learning pipeline that estimates a student's Grade Point Average (~~GPA~~) (GPA) based on academic and demographic attributes of the student. Educational institutes can use this to identify struggling students with (GPA).

[The core objective is to demonstrate a ~~robust~~ robust, clean and end-to-end machine learning workflow - encompassing data preparation, feature transformation, model training, and metric evaluation - adhering strictly to course guidelines.]

Dataset Description

The dataset utilized for this predictive task records individual academic and profile metrics for students. It contains structural columns capturing both categorical boundaries and continuous behavioral variations.

The specific variables present in the dataset are included.

Student ID: A unique identifier assigned to each individual student record.

Gender: The demographic gender of the student.

Age: The biological age of the student in years.

Study Hours Per Week: The total number of hours the student dedicated to self study and academic on weekly basis.

Attendance Rate: The overall percentage of classes attended by the student, bounded between 0% and 100%.

Major: The specific academic domain of the student.

Part-time Job: The employment status indicating whether the student balances work alongside coursework.

Extra Curricular Activity: Indicates if the student participates in organized activities outside core curricula.

(GPA): The continuous numerical Grade Point Average (GPA) achieved by the student.

($\hookrightarrow$) It serves as the target ~~to~~ variable for our regression model.

Model Building & Justification

A multiple linear Regression model was selected and implemented for this task. This model fits a linear equation to the observed data by minimizing the sum of squared residuals between the actual and predicted target values.

$$GPA = \beta_0 + \beta_1(Age) + \beta_2(Grades) + \dots$$

The target variable (GPA) is continuous and ~~numeric~~ numeric, making classification model completely inapplicable. Regression is the standard framework for this task.

$\rightarrow$ Model Evaluation

To measure performance, predictions generated from the isolated test partition were evaluated against true GPA entries using two primary evaluation metrics: Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2 Score).

The RMSE measures the average distance between the model's prediction and the actual GPA scores, a lower value indicates tighter accuracy. The R^2 score represents the proportion of variance in GPA that can be from our input features.

Conclusion

This project successfully constructed and deployed a structural machine learning pipeline to predict students (GPA). The model establishes a foundational baseline showing how academic behaviour translates into performance ~~metrics~~ metrics.

Notable limitation encountered was the small operational size, which limits the linear model's ability to map non-linear behaviour.

Future improvements could involve collecting a larger training dataset to unlock more complex non-linear estimators (such as Random Forest) and incorporating ~~the~~ historical GPA tracking variables to capture long term performance trends.

Reflection

The dataset was sourced from Kaggle, about "student-performance-data"

Completing this mini project provided me with an excellent opportunity to understand the practical realities of ~~the~~ the Machine learning workflow, ~~moving~~ moving beyond purely theoretical knowledge. I learned that designing a successful machine learning system requires spending a significant amount of time on data cleaning and feature engineering before any model training can begin. This assignment clearly demonstrated how data cleaning, normalization, model training and performance metrics are interconnected within a production pipeline. I now feel much more confident in my ability to handle raw data and deploy functional ~~models~~ models for real-world tasks.